

FORMAT ABSTRACT / FULL PAPER TEMPLATE

TYPESCRIPT

It is suggested that all papers be separated into the following sections: ABSTRACT, INTRODUCTION, MATERIALS and METHODS, RESULTS/AND DISCUSSION, ACKNOWLEDGEMENTS (if any), and REFERENCES (ABSTRAK the Malay Language translation of the Abstract should be carried out by the corresponding or main author; only in the case of foreign submissions will this translation be carried out by this Journal). INTRODUCTION, MATERIALS and METHODS, RESULTS/AND Capitalize all section titles and align them to the left. Subheadings should be typed in lowercase and flushed to the left as well. Two spaces below the section title should be typed the text. The results should be given in either table or figure format, but not both unless absolutely necessary. Captions should be written above or below the table or figure.

In general, the document should be ordered in the following order:

Title: Short and to-the-point, typed in capital and lowercase letters, bolded, and centred.

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Present address(es): centred and typed in uppercase and lowercase letters. For authors, e-mail addresses are not suggested, but are necessary for the corresponding author.

When read in isolation, the abstract (Abstrak) should be informative, short, and clear in 200 to 300 words. The words ABSTRACT and ABSTRAK should be bold and italicised. An abstract and title in English are required for papers written in Malay and Arabic.

Keywords (Kata kunci): Five or six terms that reflect the paper's overall concepts should be chosen as keywords. Keywords and Kata kunci are two terms that are used interchangeably.

Bibliographical ~~Supplementary material~~

~~Characterization of nanoprecipitated PET nanoplastics by ¹H NMR and impact of residual ionic surfactant on viability of human primary mononuclear cells and hemolysis of erythrocytes~~

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~~Preparation of PET NPs~~

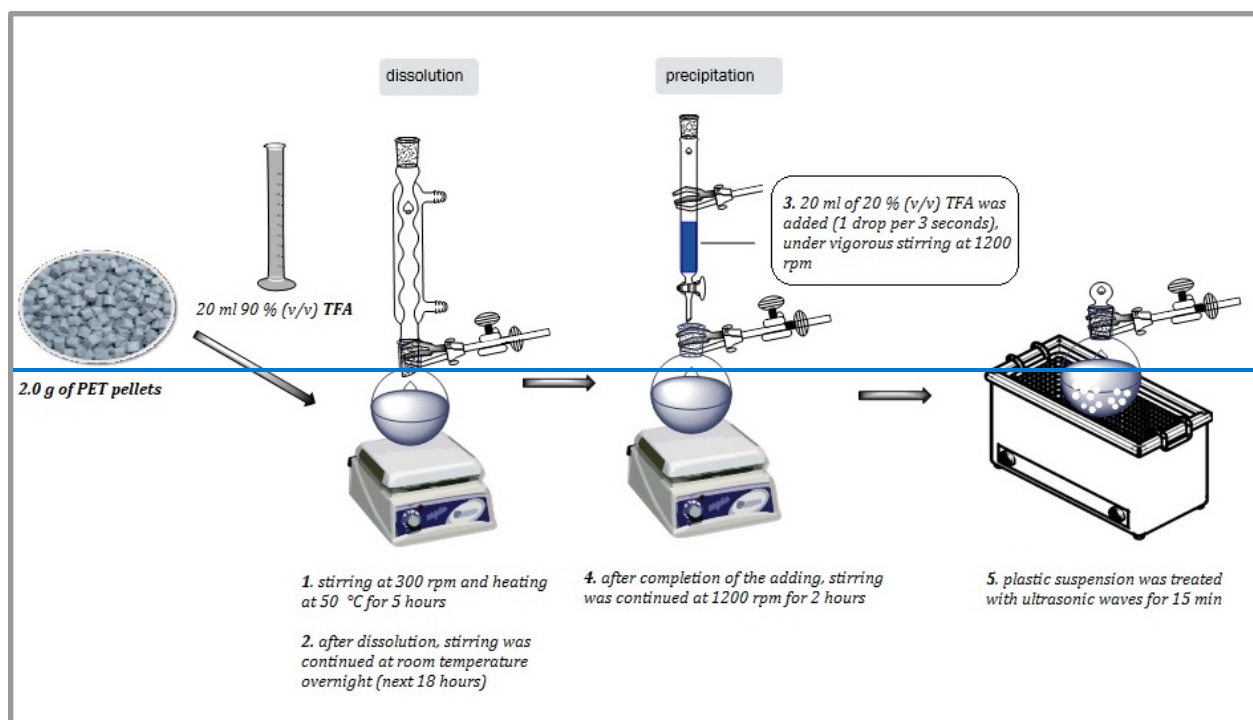


Figure S1. Overall scheme of nPET production procedure. First, 2.0 g of PET pellets was dissolved in 20.0 mL of TFA in Milli-Q water (v/v) with stirring at 300 rpm for five hours at 50 °C and next 18h at room temperature. Then, 20 mL of 20% TFA in Milli-Q water (v/v) was added dropwise during 110 min (1 drop of 10 μ L per 3 s), under vigorous stirring at 1200 rpm using a dropping funnel. Stirring of suspension was continued for an additional 2 hours, before being sonicated in an ultrasonic bath for 15 min.

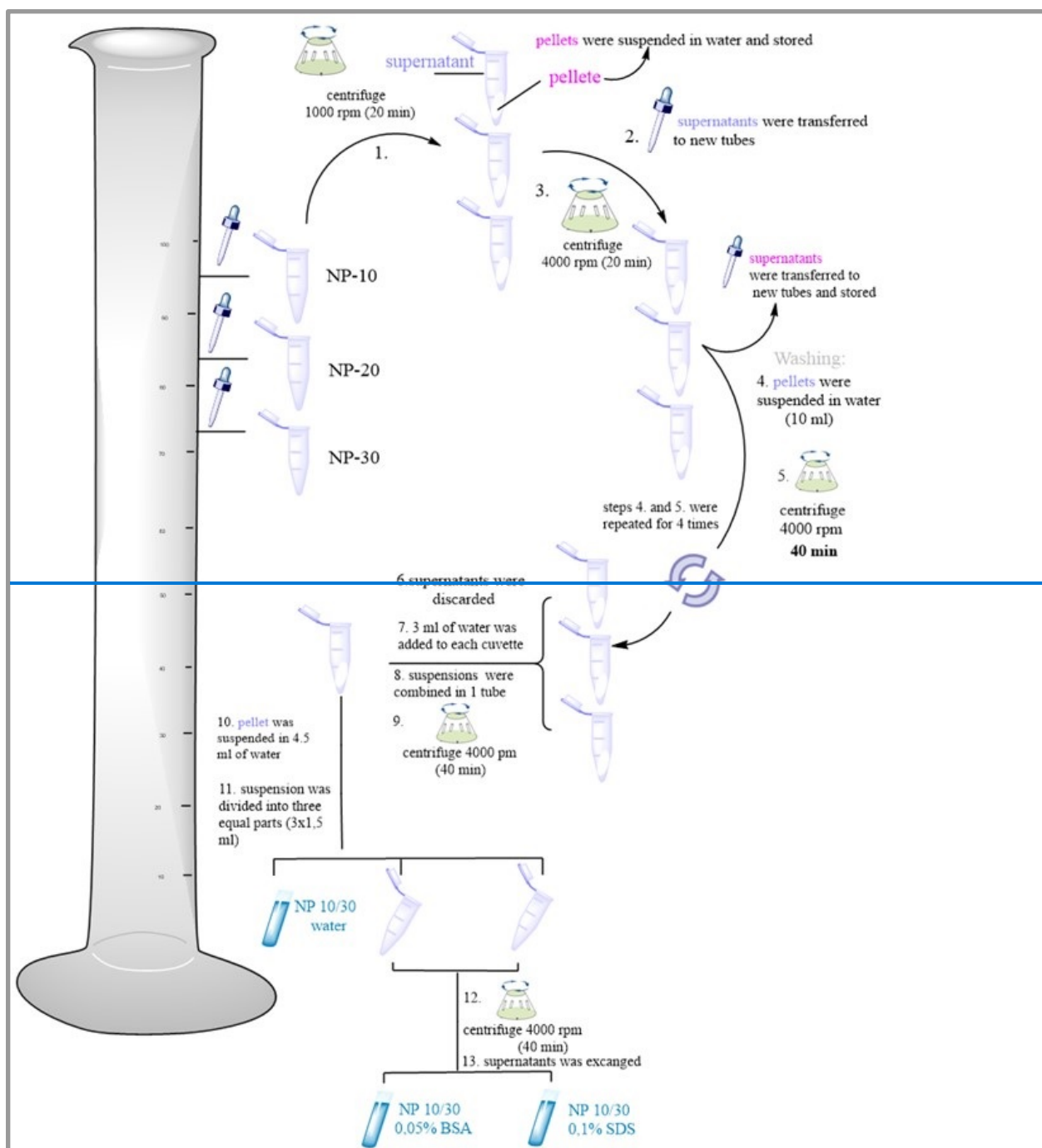


Figure S2. Schematic illustration of washing and size separation processes of the first three fractions (each 10 mL), denoted NP-10, NP-20 and NP-30.

Characterization of PET NPs

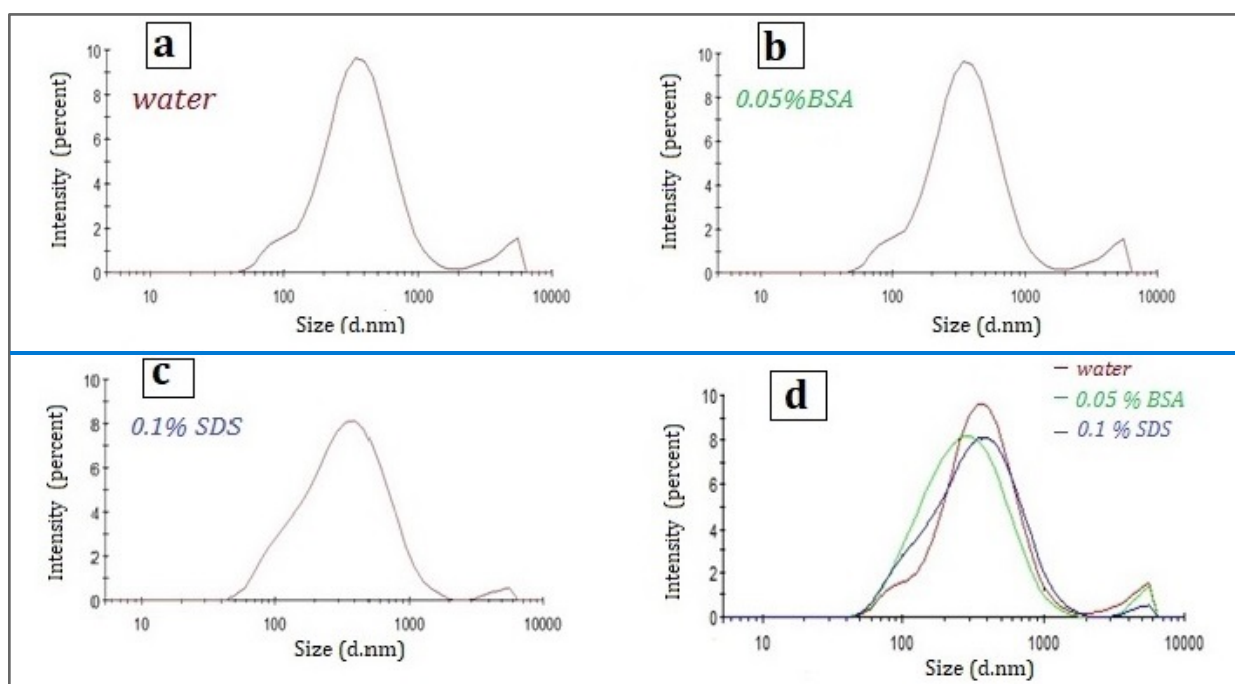


Figure S3. Size distributions (by intensity percentage) of NPs washed suspended in different dispersants: water, BSA (0.05%) and SDS (0.1 %) are presented. Size distributions are shown for NPs obtained from combined fractions: NP-10, NP-20 and NP-30.

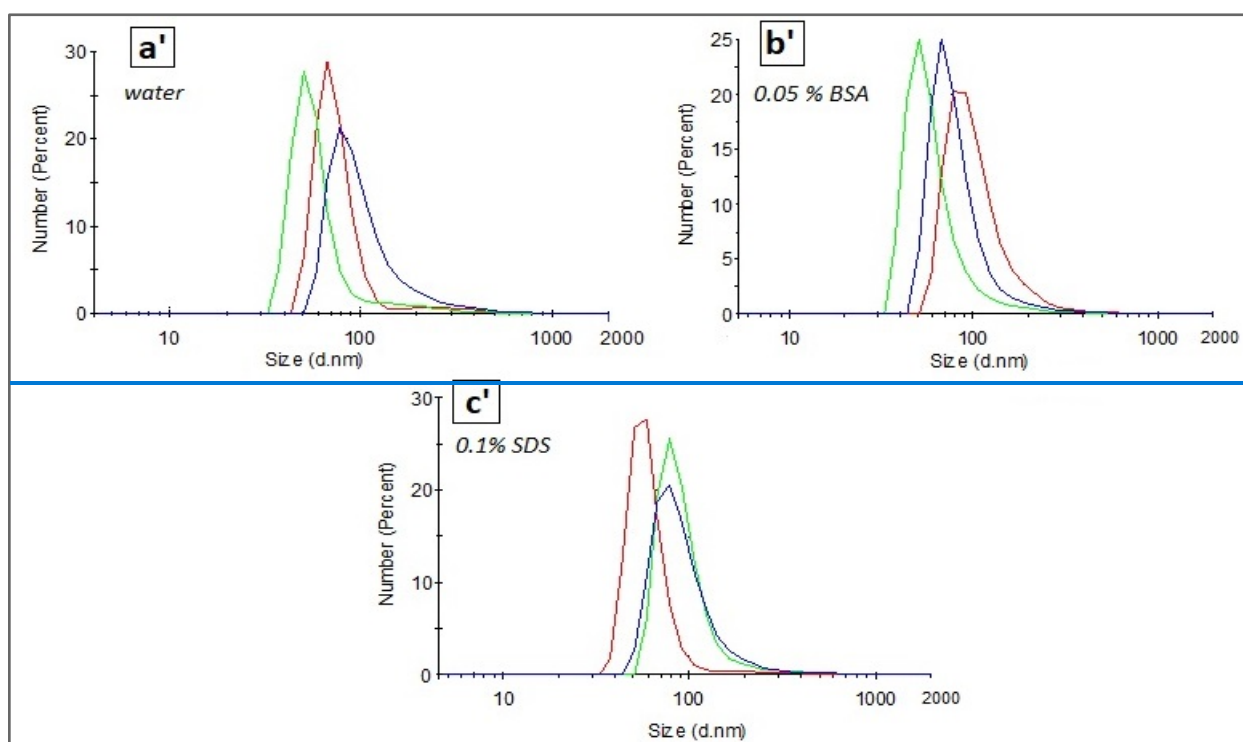


Figure S4. Size distributions (by number percentage) of NPs washed suspended in different dispersants: water, BSA (0.05%) and SDS (0.1 %) are presented. Distributions are shown for NPs obtained from combined fractions: NP-10, NP-20 and NP-30. Additionally multiple consecutive measurements (lines with different colors) are presented.

Table S1. Hydrodynamic diameter (d.nm) and polydispersity index of NPs washed obtained from combined fractions: NP 10, NP 20 and NP 30 (denoted NP 10+NP 20+NP 30); NP 40, NP 50 and NP 60 (denoted NP 40+NP 50+NP 60); NP 70, NP 80 and NP 90 (denoted NP 70+NP 80+NP 90) dispersed in different dispersants: water, 0.05% BSA and 0.1% SDS.

	dispersant	NPS washed		
		NP-10+NP-20+NP-30	NP-40+NP-50+NP-60	NP-70+NP-80+NP-90
Z-Average (d.nm)	water	311.7 ± 4.4	182.3 ± 3.4	262.5 ± 9.6
	0.05% BSA	243.5 ± 1.3	264.6 ± 7.2	230.7 ± 1.1
	0.1 % SDS	265.7 ± 5.9	256.8 ± 3.3	215.6 ± 1.6
Polydispersity index	water	0.375 ± 0.026	0.309 ± 0.034	0.540 ± 0.073
	0.05% BSA	0.352 ± 0.037	0.271 ± 0.010	0.269 ± 0.011
	0.1 % SDS	0.351 ± 0.041	0.336 ± 0.026	0.260 ± 0.050

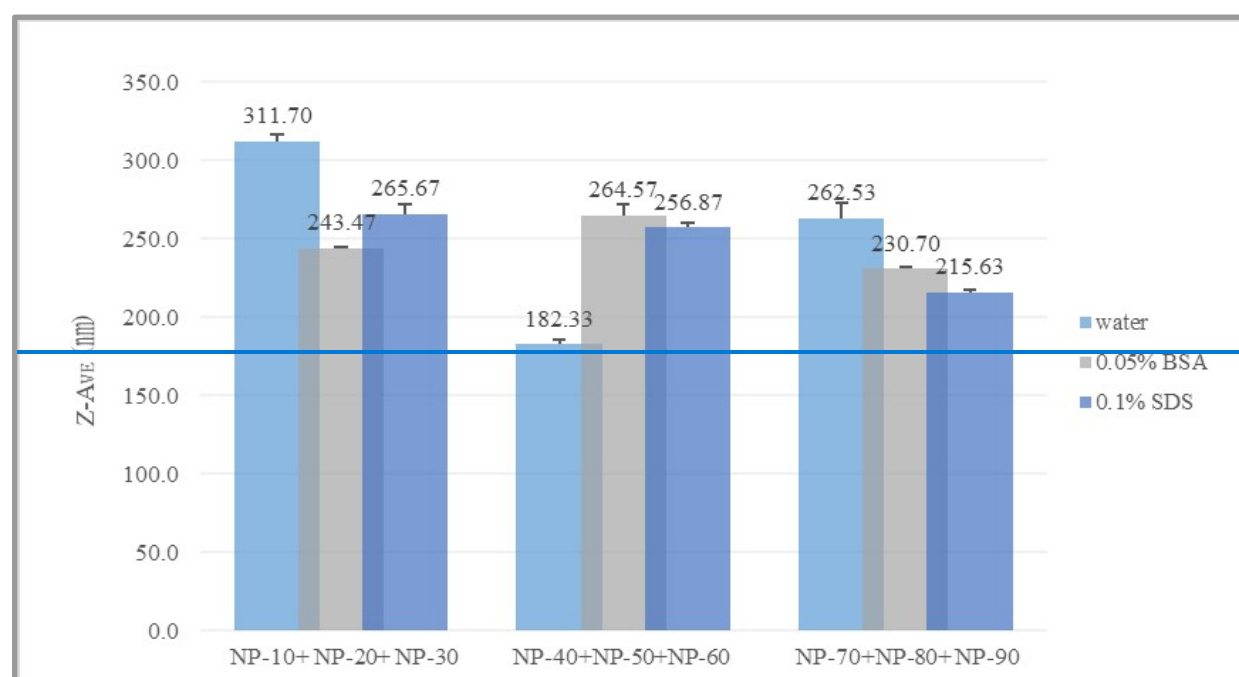


Figure S5. Hydrodynamic diameter (d.nm) of NPs washed obtained from combined fractions: NP 10, NP 20 and NP 30 (denoted NP 10+NP 20+NP 30); NP 40, NP 50 and NP 60 (denoted NP 40+NP 50+NP 60); NP 70, NP 80 and NP 90 (denoted NP 70+NP 80+NP 90) dispersed in different dispersants: water, 0.05% BSA and 0.1% SDS.

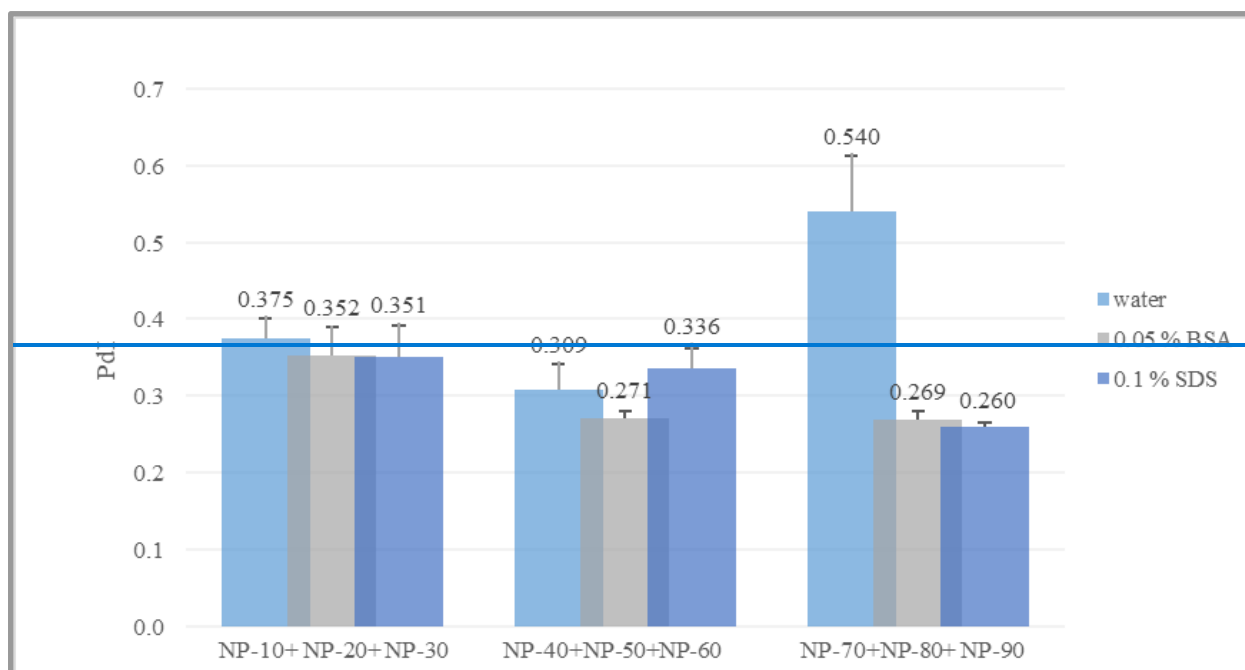


Figure S6. Polydispersity index of NPs washed obtained from combined fractions: NP 10, NP 20 and NP 30 (denoted NP 10+NP 20+NP 30); NP 40, NP 50 and NP 60 (denoted NP 40+NP 50+NP 60); NP 70, NP 80 and NP 90 (denoted NP 70+NP 80+NP 90) dispersed in different dispersants: water, 0.05% BSA and 0.1% SDS.

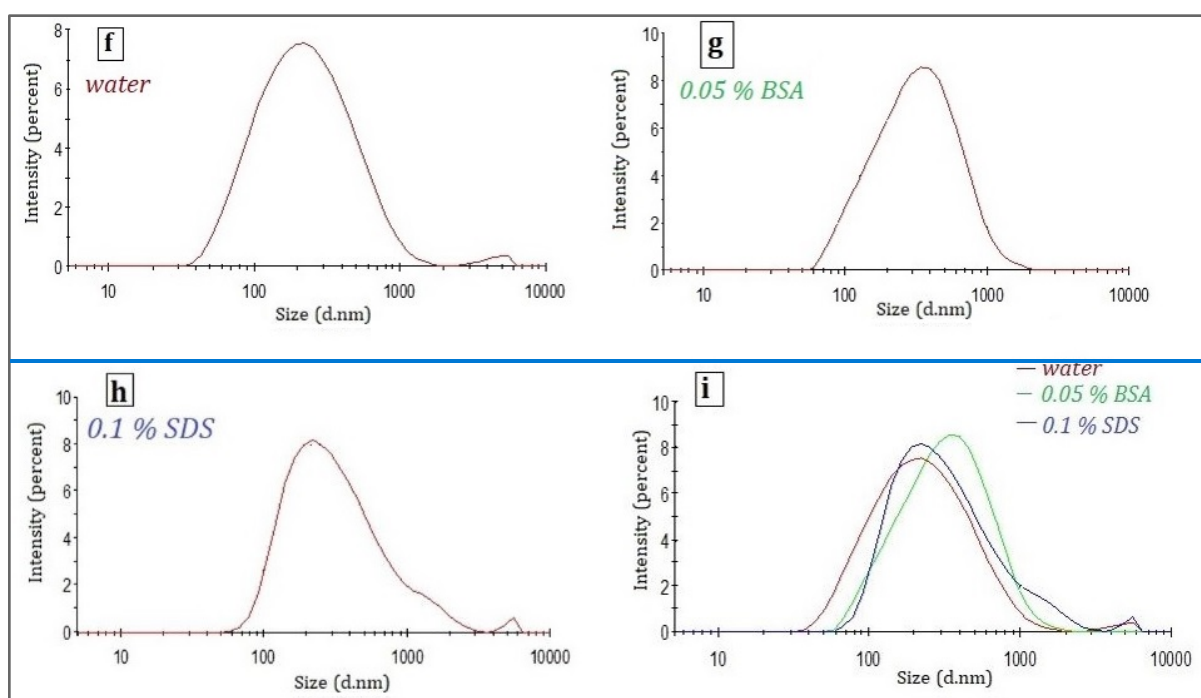


Figure S7. Size distributions (by intensity percentage) of NPs washed suspended in different dispersants: water, BSA (0.05%) and SDS (0.1 %) are presented. Size distributions are shown for NPs obtained from combined fractions: NP 40, NP 50 and NP 60.

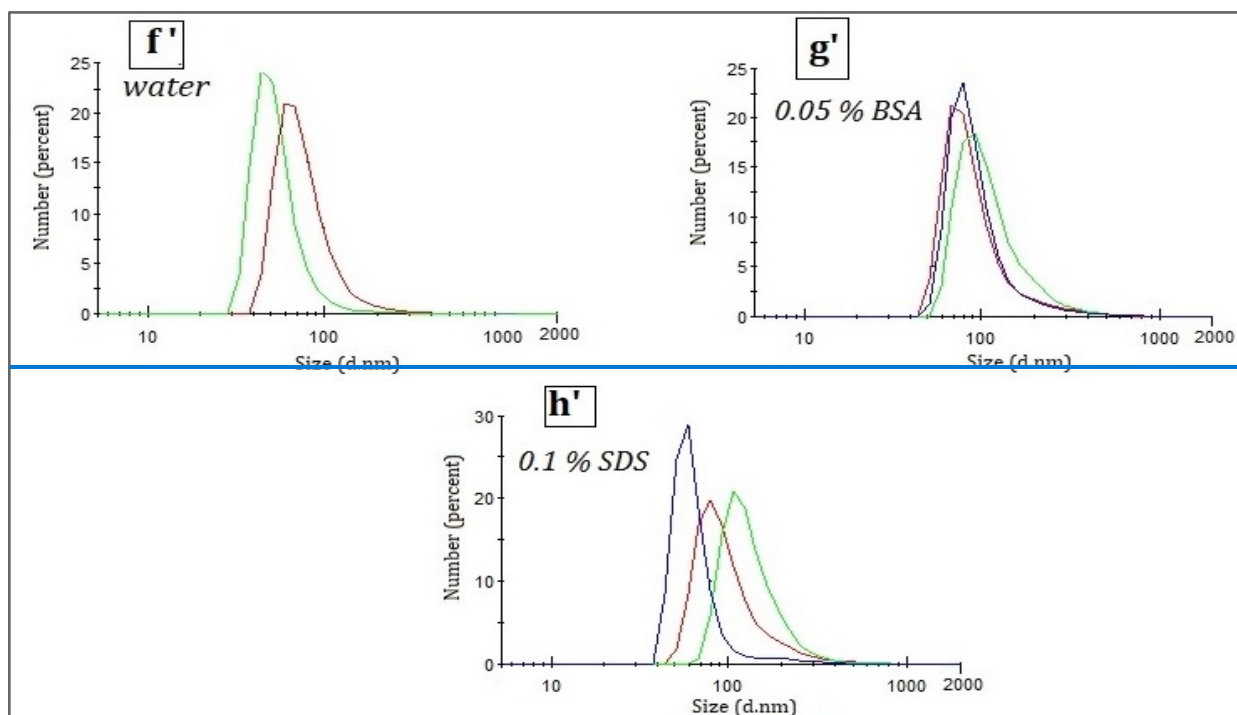


Figure S8. Size distributions (by number percentage) of NPs washed-suspended in different dispersants: water, BSA (0.05%) and SDS (0.1 %) are presented. Distributions are shown for NPs obtained from combined fractions: NP 40, NP 50 and NP 60. Additionally multiple consecutive measurements (lines with different colors) are presented.

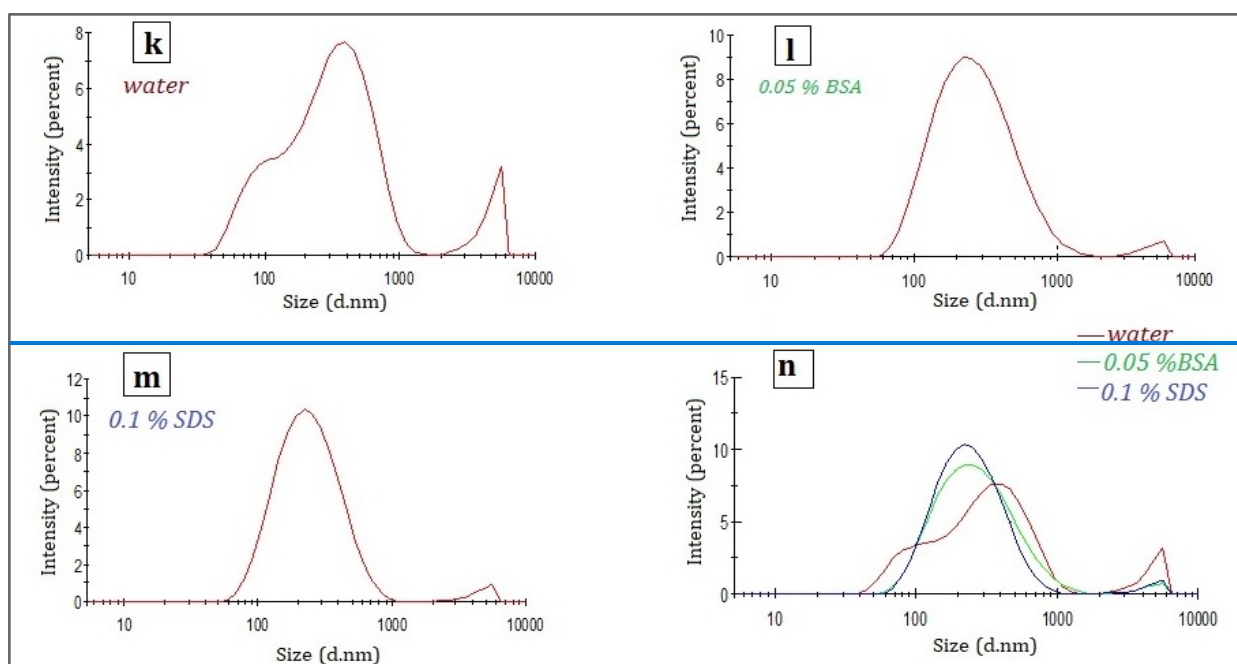


Figure S9. Size distributions (by intensity percentage) of NPs washed-suspended in different dispersants: water, BSA (0.05%) and SDS (0.1 %) are presented. Size distributions are shown for NPs obtained from combined fractions: NP 70, NP 80 and NP 90.

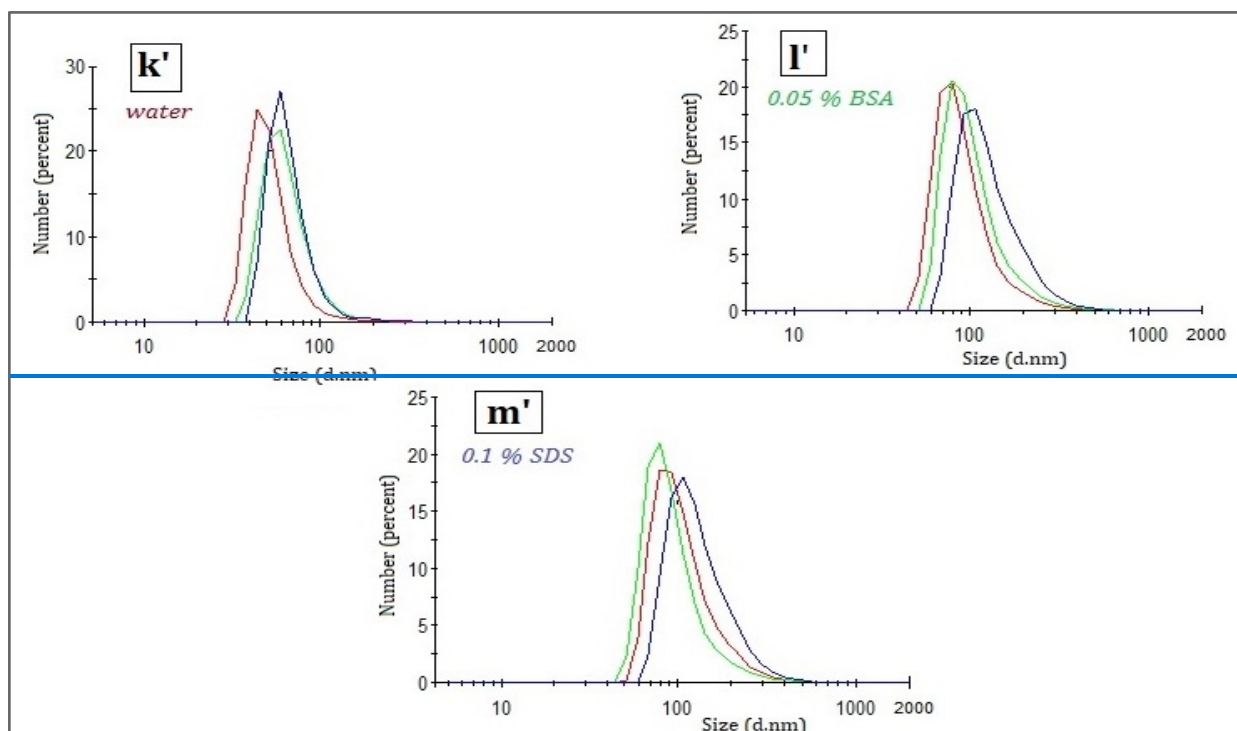


Figure S10. Size distributions (by number percentage) of NPs washed suspended in different dispersants: water, BSA (0.05%) and SDS (0.1 %) are presented. Distributions are shown for NPs obtained from combined fractions: NP-70, NP-80 and NP-90. Additionally multiple consecutive measurements (lines with different colors) are presented.

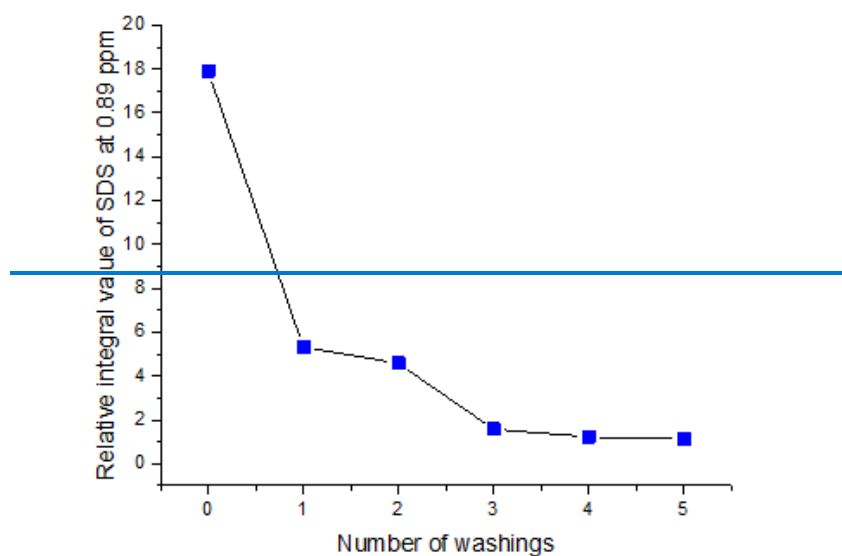


Figure S11. Relative quantification of SDS contamination in relation to PET signal in NMR.

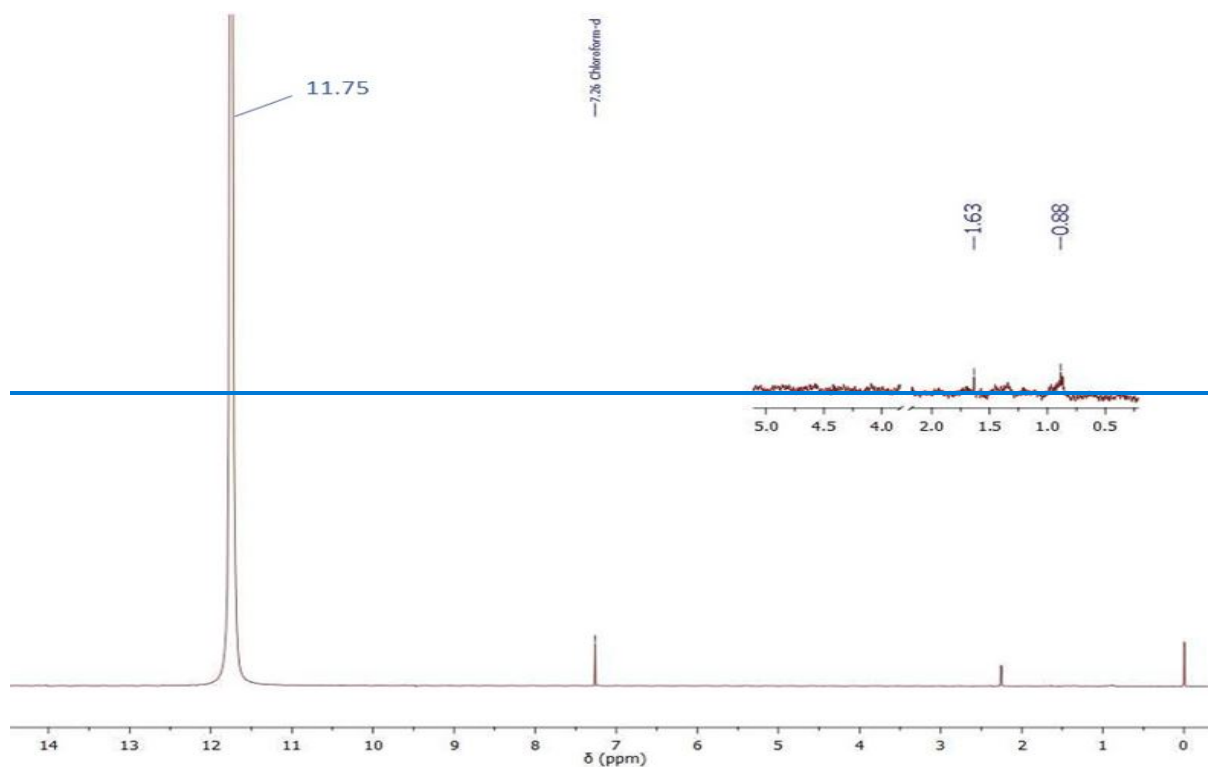


Figure S12. ^1H NMR spectrum (400 MHz) of trifluoroacetic acid (TFA) recording with addition of deuterated chloroform (CDCl_3) in the ratio 4:1 (v.v, respectively). Region importance for process monitoring of washing process of NPs (0.5–2.0 ppm and 4.0–5.0 ppm) is zoomed. Signal at 11.75 ppm originates from TFA; signals with chemical shifts values 0.88 and 1.63 ppm indicate the presence of impurities in very low concentrations.

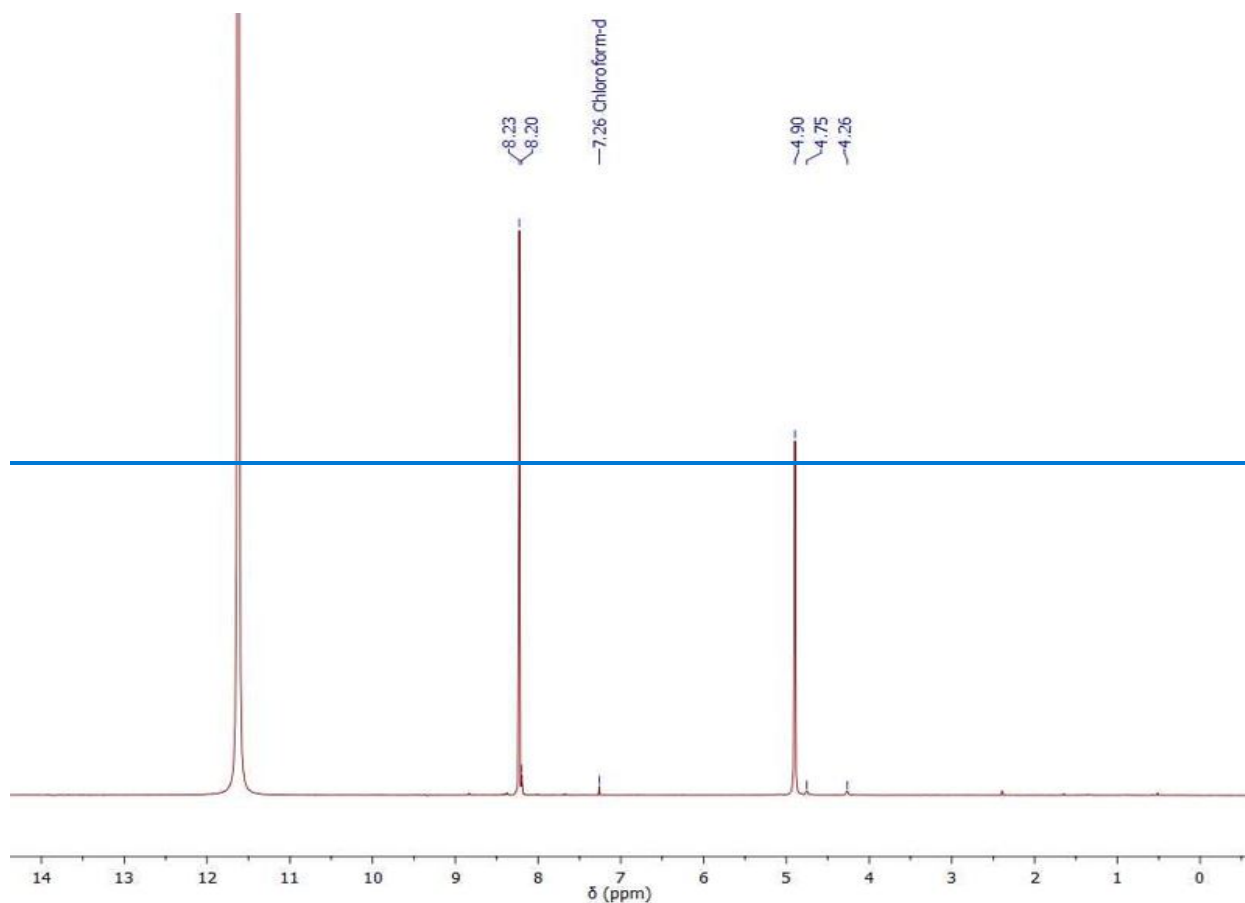


Figure S13. ^1H NMR spectrum (400 MHz) of PET pellet (starting material) recorded in the mixture of TFA and CDCl_3 in the ratio 4:1 (v/v). The use of this mixture containing the fluorinated solvent is advantageous, since it allows room temperature spectra [1]. Characteristic signals at 8.20–8.23 (aromatic protons), 4.90 (internal methylene groups), and low intensity signals at 4.75 and 4.26 ppm (the methylene protons adjacent to the hydroxyl end functions) are consistent with reported literature data [2,3].

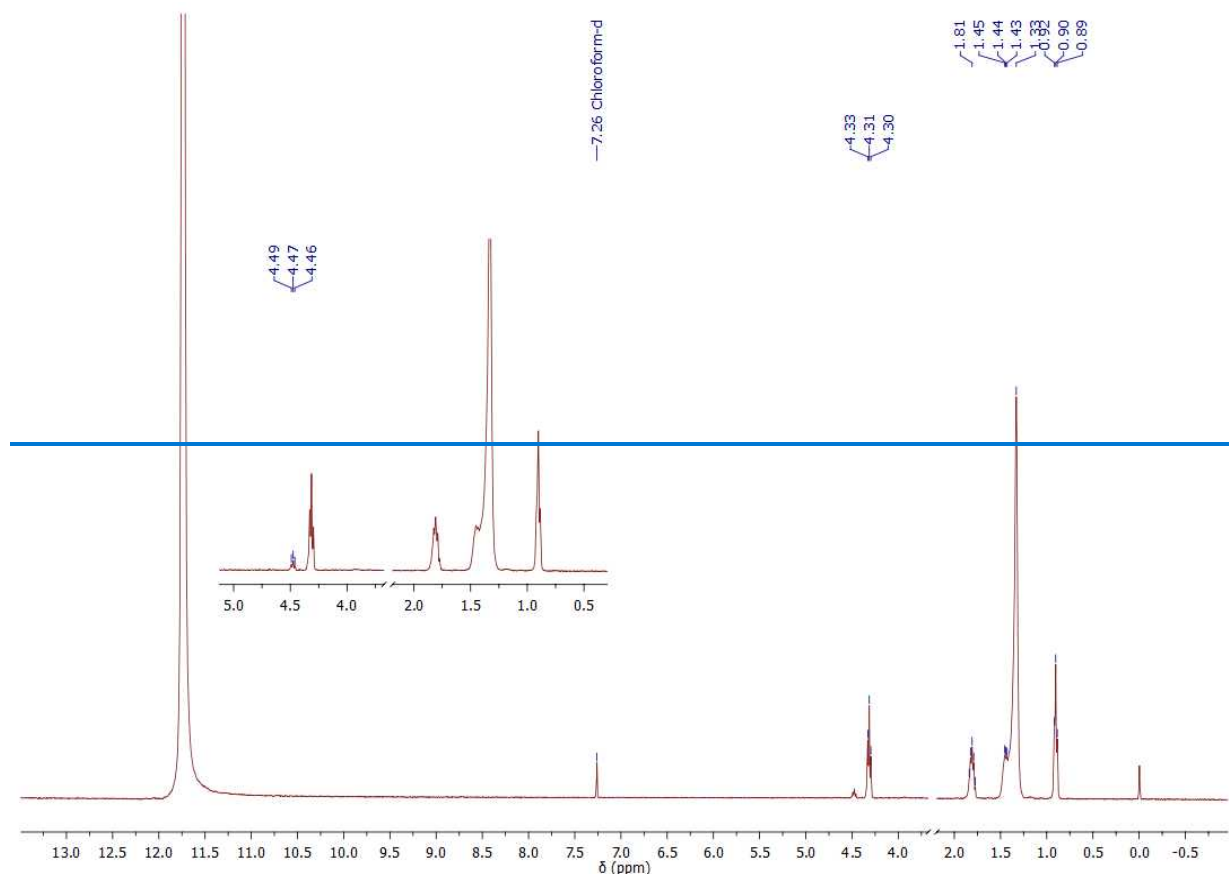


Figure S14. ^1H NMR spectrum (400 MHz) of sodium dodecyl sulfate (SDS) recorded in the mixture of TFA and CDCl_3 in the ratio 4:1 (v/v). The peaks at regions 0.89–0.92, 1.33–1.45, 1.81 and 4.30–4.33 ppm originate from SDS. In the magnified part of the spectrum a low intensity signal with chemical shift of 4.46–4.49 ppm is visible. The intensity of this impurity decreases during the washing process of NPs from SDS; already after third washing, this peak is not visible in the NMR spectrum (Figure 1B).

References:

The list of references should be alphabetized by the first author's surname. Publication(s) in the same year by the same author(s) should be noted as 2000a, 2000b, and so on. Journal titles should be capitalised and referenced in full. The numbers for the volume and issue should be bold.

Except when the author is specified, e.g., and the study of Ross and Ross (1990) and Salisbury et al. (1995), references listed in the text should be included in parenthesis, e.g. (Mohammad 1992; Musa and Hassan 1993; Ho et al. 1994).

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Journal citation: [Ali, A. M., Hamid, M. & Yusoff, K. 1996. Production of monoclonal antibodies against Newcastle disease virus. *Asia Pacific Journal of Molecular and Biotechnology* 4: 236-242.](#)

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1. ~~Book:~~ [Airey, B.; Moad, G.; Van Diepen, G.; Willing, I.; Cook, W. D. Characterization of poly\(ethylene terephthalate\) and poly\(ethylene terephthalate\) blends. *Polymer*, **1997**, 38\(12\), 3035-3043, doi:10.1016/s0032-3861\(96\)00872-5.](#)
2. [El Mejjati, A.; Harit, T.; Riahi, A.; Khiari, R.; Bouabdallah, I.; Malek, F. Chemical recycling of poly\(ethylene terephthalate\). Application to the synthesis of multiblock copolyesters. *EXPRESS Polym. Lett.* **2014**, 8, 544553, doi: doi:10.3144/expresspolymlett.2014.58.](#)

[Saint-Loup, R.; Jeanmaire, T.; Robin, D. 2010. Logo design love: A guide to creating iconic brand identities. Berkeley: New Riders.](#)

[Lateh, H.M. 2018. Undang-undang Tubuh Kerajaan Johor 1312H/1895. Bangi: Penerbit Universiti Kebangsaan Malaysia.](#)

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[American Psychological Association. 2010. Concise rules of APA style \(6th ed.\). Washington, DC: Author.](#)

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Book chapter: [Ahmad, B., Hashim, Z., Wood, A. K., Mat, N. & Rashid, M. K. A. R. 2002. Speciations of organochlorine pesticides in sediment of coastal water of the Muda rice granary area. p. 141-52. In. B. M. Nashriyah, N. K. Ho, B. S. Ismail, A. B. Ali, K. Y. Lum & M. B. Mansor \(eds.\), Sustainable Rice Production in Malaysia Beyond 2000. \(pp. 39-50\). Malaysia: MINT-MADA.](#)

[Hawthorne, J., Kelsch, A., & Steen, T. 2010. Making general education matter: Structures and strategies. In. C. M. Wehlburg \(Ed.\), Integrated general education \(2nd ed.\) \(pp.23-34\). San Francisco, CA: Jossey-Bass.](#)

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Proceedings: Zakaria, M. B., Jais, M. J., Wan-Yaacob Ahmad, Mohd Rafidi Othman & Zainal Abidin Harahap. 2002. Penurunan kekeruhan EIMS oleh koagulan konvensional dan kitosan. In. Prosiding Seminar Kimia Bersama UKM ITB. 16-17 July 2002, Melaka, Malaysia. p. 487-492.

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Presentations: Mat B. Zakaria & Md. Pauzi Abdullah. 1996. Potential of chitosan in the food industries. Presented at First National Technology Connection Seminar. 14-15 November 1996, Kuala Lumpur, Malaysia. 7 pp.

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Report: Nashriyah Mat & Norimah Yusof. 2005. Analysis of Prawn Tissue Samples Derived from Mini Study on the Impact of Coal Dust to Prawn Fry. **MINT/L/2005/38(S)** (Research Project Agreement MINT-TNBJ: Code No: **UPP/2005/(03)**). Malaysian Nuclear Agency, Bangi, Kajang, Selangor, Malaysia. 108 pp.

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Thesis and Dissertation:

Embong, R. 2011. The concept of Islamic integrated curriculum and its implications on Islamic schools in Malaysia. Ph.D Thesis, International Islamic University Malaysia, Kuliyyah of Education, Gombak, Malaysia.

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Patent: Andress, Jr. H. J. 1974. German Patent 2, 413, October.

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Standard: Japanese Industrial Standard. 1975. Method of Ultrasonic Manual Testing and Classification of Test Results for Welds, JIS Z 3060.

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Internet: JAIF, Inc. 2004. 6th FNCA Coordinators Meeting in Japan, <http://www.fnca.jp>, retrieved 1st June 2005.

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Acknowledgements

This should be included whenever it is deemed appropriate, as a separate section before the Reference index section.

- ~~3. J.J.; Boutevin, B. Synthesis of (polyethylene terephthalate/polycaprolactone) copolyesters. *Polymer*, **2003**, *44*, 3437-3449, doi: 10.1016/S0032-3861(03)00257-X.~~